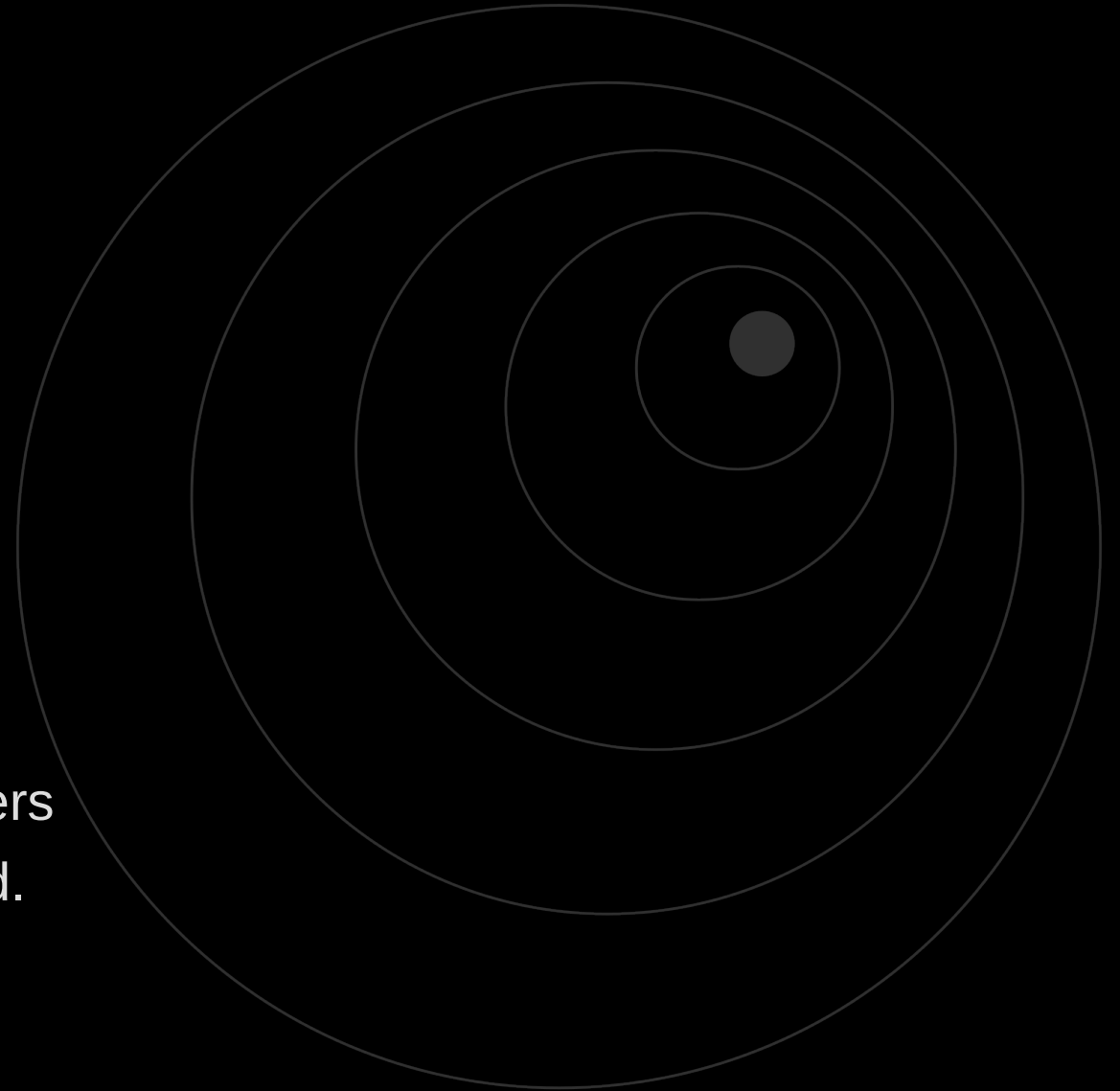


R ROTO

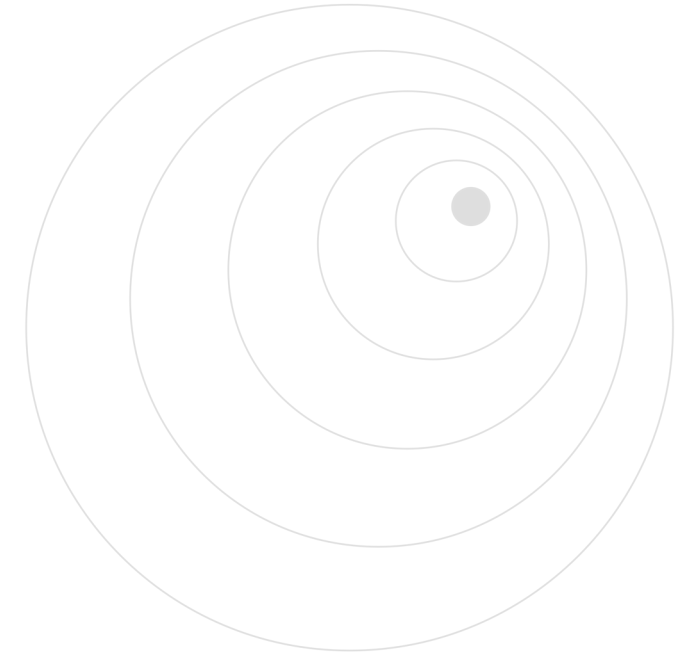
Aerial data for quarries. Weekly stockpile numbers instead of a once-a-month surveyor walk-around.



PROBLEM

A quarry is blind between surveys

The staff surveyor walks the yard with a total station once a month. It takes half a day. Everything between those visits, short-loading, shrinkage, plain theft, piles up unseen until the next survey.



1x

per month is how often most yards get measured

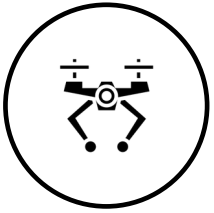
400–700K

KZT monthly salary of a mine surveyor (estimate, hh.kz vacancies)

+15.9%

construction growth in Kazakhstan in 2025, quarries feed it on monthly numbers

Drone scan in, volume report out

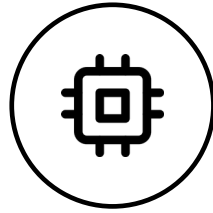


Fly

A 249 g drone flies an automated route over the stockyard. Twenty minutes per yard.

under 5%

target error vs manual survey, proven on a reference pile in week one



Process

Photogrammetry pipeline on open-source WebODM builds a 3D model and computes each pile's volume.

under 500K KZT

one-off hardware cost, bought in Almaty within 3 days



Report

Deltas against the previous scan, full stock dynamics. A subscription, 2 scans a month.

zero permits

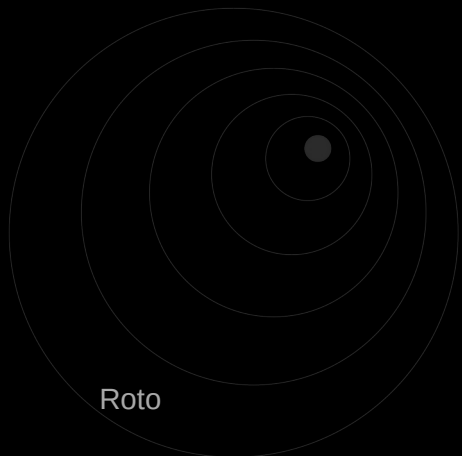
under the 250 g threshold, outside populated areas, open category

28

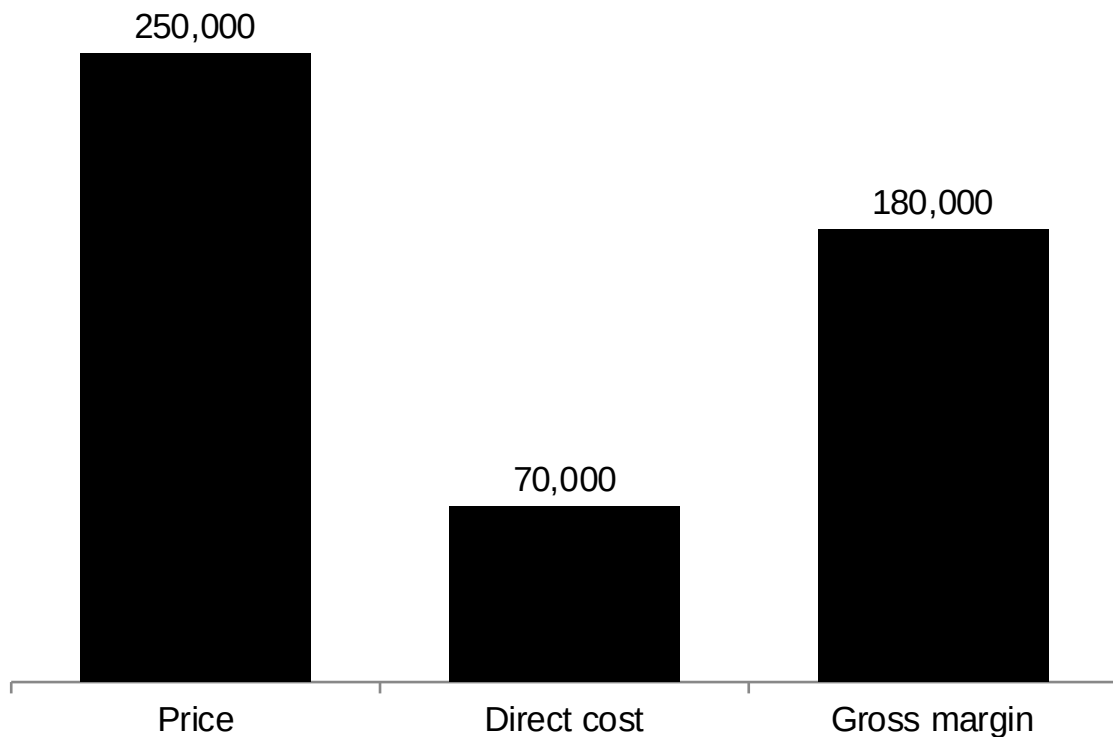
named buyers within a day's drive

The regional catalog lists 28 quarries and sand-and-stone producers around Almaty. Asphalt and concrete plants with bulk yards are the same sale. The buyer is private, no tenders, one person says yes.

2027. Second vertical on the same stack, construction progress monitoring for Almaty developers.



Subscription with service margins



72% gross margin

at a 250 000 KZT subscription, direct cost is about 70 000 KZT per client-month.

2 clients to break even

fixed costs are covered by the second subscriber.

30–60K KZT CAC

a free demo scan on the client's own yard, paid back by the first month.

No cash gap

prepaid monthly, in tenge both ways. The only frozen cash is one drone, bought once.

The rival is a habit, not a company

In-house surveyor

The status quo. We complement him, he comes monthly, we fill the weeks between.

Contract firms, freelancers

One-off surveys from ~150K KZT. No continuity, no stock dynamics between visits.

Propeller, DJI Terra

International software proves the method works. Nobody packages it as a subscription for quarries in our region.

The empty niche is a question

Untapped market or absent demand. Our first 15 customer interviews are designed to tell those apart.

From reference pile to paying clients

W1–2



Buy the drone, build a reference pile, publish measurement error against manual survey.

W3–4



5 free demo scans across the call list, 10 customer interviews.

W5–8



Price set from interviews, first paid invoice, processing automated.

W9–12



2 paying subscriptions, 2 LOIs, a discrepancy case. Demo Day.

Kill threshold set in advance. Fewer than 2 paying organizations by November 15 and we call the hypothesis failed, with the data published either way.

TEAM

The team that signed a metro

Amangeldi Bekaidar

Co-founder and CEO · full-time

Sales, negotiations, drone pilot on site visits.
Based in Almaty.

Myrzabay Beknurtas

Founder and CTO · full-time

Photogrammetry pipeline, measurement accuracy. Built LocalQuakeAlert (ESP32) solo.

Bakhtiyar Merey

Communications · part-time

Design, content, customer interviews across both projects.

1st place

TURAQTY JOL 7.0

Signed LOI

Almaty Metro, EcoStep pilot

Invited

TechCrunch Disrupt 2026

Same three people behind EcoStep, with a split calendar between the two projects.



THE ASK

A tracker who has sold to heavy industry, pressure on our numbers, and a Demo Day deadline

We bring the checked idea, the hardware budget and the call list. The program brings the discipline. By December 6 that adds up to 2 paying quarries or an honest dataset on why not.